

The Sharp $\text{cov}(\mathcal{N})$ Threshold for Three Families of Pathological Nets

Math discovery candidate partial result

August 17, 2026

1 Status and source questions

This packet gives a candidate substantial partial answer to Questions Q1–Q3 in D. L. Rodríguez-Vidanes, *Lineability on nets and uncountable sequences of functions in measure theory*, arXiv:2306.03862v1 [1]. The questions occur in Section 4, printed page 26:

4. FINAL REMARKS

In Section 2 we were interested in finding any directed set (that is not a cardinal number) in which there was an algebraic structure such as a cone or an algebra. In any case, not only we tried to find the most optimal structure (for instance, in the case of algebras, we were trying to find algebraically independent nets) but we were also studying the largest possible dimension of these structures. Moreover, notice that we tried to find the smallest possible cardinality of a directed set such that the family of nets in this paper contained a certain algebraic structure. Keeping the above in mind it is natural to ask the following questions:

- (Q1) Let $1 \leq \mu \leq 2^{\mathfrak{c}}$ be a cardinal number. For which directed sets $\mathcal{A}_\mu$ does there exist a positive cone of dimension μ contained in $\mathcal{NM}_\mathcal{A} \cup \{0\}$, $\mathcal{NF}_\mathcal{A} \cup \{0\}$ and $\mathcal{ND}_\mathcal{A} \cup \{0\}$ in $(\mathbb{R}^{[0,1]})^\mathcal{A}$? What is the smallest possible cardinality of $\mathcal{A}_\mu$?
- (Q2) Let $1 \leq \mu \leq 2^{\mathfrak{c}}$ be a cardinal number. For which directed sets $\mathcal{A}_\mu$ does there exist a cone of dimension μ contained in $\mathcal{ND}_\mathcal{A} \cup \{0\}$ in $(\mathbb{R}^{[0,1]})^\mathcal{A}$? What is the smallest possible cardinality of $\mathcal{A}_\mu$?
- (Q3) Let $1 \leq \mu \leq 2^{\mathfrak{c}}$ be a cardinal number. For which directed sets $\mathcal{A}_\mu$ are the families $\mathcal{AN}_\mathcal{A}$ (in $(\mathbb{R}^\mathbb{R})^\mathcal{A}$), $\mathcal{ND}_\mathcal{A}$ and $\mathcal{ANM}_\mathcal{A} \setminus \mathcal{ND}_\mathcal{A}$ (in $(\mathbb{R}^{[0,1]})^\mathcal{A}$) strongly μ -algebraable? What is the smallest possible cardinality of $\mathcal{A}_\mu$?

The result completely determines the smallest-cardinality clause of Q1 and Q2 for every dimension requested there, and of Q3 up to dimension $\mathfrak{c}$. It does not classify all directed sets.

2 Definitions

Write $\mathcal{N}$ for the ideal of Lebesgue-null subsets of $[0, 1]$, and $\text{cov}(\mathcal{N})$ for the least cardinality of a subfamily of $\mathcal{N}$ covering $[0, 1]$. Let $\mathcal{A}$ be a directed set. We use the source paper's families:

- $\mathcal{NM}_\mathcal{A}$ consists of a.e.-increasing nets $(f_a)_{a \in \mathcal{A}}$ of nonnegative measurable functions which converge pointwise a.e. to an integrable f , but for which monotone convergence of the integrals fails.
- $\mathcal{NF}_\mathcal{A}$ consists of nets of a.e.-nonnegative measurable functions for which

$$\int \liminf_{a \in \mathcal{A}} f_a > \liminf_{a \in \mathcal{A}} \int f_a,$$

with the pointwise net $\liminf$ integrable.

- $\mathcal{ND}_{\mathcal{A}}$ consists of nets (f_a) , dominated a.e. at every index by one integrable function, which converge pointwise a.e. to an integrable f , but for which $\int |f_a - f| \not\rightarrow 0$.

All integrals below are over $[0, 1]$.

3 Proof intuition

The usual sequence proofs conceal a countability step. In the Lebesgue measure algebra, even an uncountable essential union or essential supremum is determined by a countable subfamily. If a dominated net has large L^1 -error arbitrarily far out, every index tail therefore has a positive-measure essential union of large-error sets, and the essential tail-limsup still has positive measure. Pointwise convergence forces each point of that limsup into a null exceptional set attached to some index. Thus the index set supplies a null cover of a positive-measure set, requiring at least $\text{cov}(\mathcal{N})$ indices.

For monotone nets, compare the literal pointwise limit with the essential supremum of all terms. For Fatou, compare the literal tail infima with the essential tail infima. The same null-cover obstruction makes these objects agree in the direction needed for the classical integral inequalities.

4 Measure-algebra lemmas

Lemma 1 (Positive measurable sets have covering number $\text{cov}(\mathcal{N})$). *If $E \subset [0, 1]$ is measurable and $\lambda(E) > 0$, and $E \subset \bigcup_{\xi < \kappa} N_\xi$ with every $N_\xi \in \mathcal{N}$, then $\kappa \geq \text{cov}(\mathcal{N})$.*

Proof. Work modulo 1 on the circle. There are countably many translates of E whose union is conull. Indeed, if R is the uncovered measurable remainder, averaging over translations t gives

$$\int \lambda(R \cap (E + t)) dt = \lambda(R)\lambda(E).$$

Choose a translate covering at least this proportion of R , and iterate; the remainder decreases geometrically to a null set.

If $\kappa < \text{cov}(\mathcal{N})$, each selected translate of E is covered by κ translates of the N_ξ 's. These null sets, together with the null remainder, form a cover of the circle by at most $\kappa \cdot \aleph_0 < \text{cov}(\mathcal{N})$ null sets, contradicting the definition of $\text{cov}(\mathcal{N})$. $\square$

Lemma 2 (Countable determination). *Every essential union or essential intersection of an arbitrary family of Lebesgue-measurable sets is equal modulo null sets to the corresponding operation on a countable subfamily. Consequently, every essential supremum or essential infimum of an arbitrary family of measurable extended-real functions is determined a.e. by a countable subfamily.*

Proof. For a family $\mathcal{E}$ of measurable sets, let

$$s = \sup \left\{ \lambda \left(\bigcup \mathcal{F} \right) : \mathcal{F} \subset \mathcal{E} \text{ finite} \right\}.$$

Choose finite $\mathcal{F}_n \subset \mathcal{E}$ for which the displayed measures tend to s , and let $U = \bigcup_n \bigcup \mathcal{F}_n$. Then $\lambda(U) = s$. For every $E \in \mathcal{E}$, one has $\lambda(E \setminus U) = 0$, since otherwise $\lambda(U \cup E) > s$. Hence U is the essential union. Complements give essential intersections.

For functions, apply the set statement to the rational superlevel sets $\{h_i > q\}$, $q \in \mathbb{Q}$, and take the union of the countably many resulting countable index sets. Essential infima follow by reversing the inequalities. $\square$

Lemma 3 (A dominated bad net needs $\text{cov}(\mathcal{N})$ indices). *Let $(h_a)_{a \in \mathcal{A}}$ be measurable functions such that $0 \leq h_a \leq G$ a.e. for every a , where $G \in L^1[0, 1]$, and suppose $h_a \rightarrow 0$ pointwise a.e. If $\int h_a \not\rightarrow 0$, then*

$$|\mathcal{A}| \geq \text{cov}(\mathcal{N}).$$

Proof. There is an $\varepsilon > 0$ such that every index tail contains a b with $\int h_b \geq \varepsilon$. Choose $M > 0$ so that $\int_{\{G > M\}} G < \varepsilon/4$, put $\delta = \varepsilon/4$, and, for each such large b , set

$$E_b = \{x : G(x) \leq M, h_b(x) \geq \delta\}.$$

The estimate

$$\varepsilon \leq \int h_b \leq \frac{\varepsilon}{4} + \delta + M\lambda(E_b)$$

shows that $\lambda(E_b) \geq \eta := \varepsilon/(2M) > 0$.

For $a \in \mathcal{A}$, let U_a be the essential union of all E_b with $b \geq a$ and $\int h_b \geq \varepsilon$. Every tail contains such a b , so $\lambda(U_a) \geq \eta$. The family (U_a) is down-directed. Let

$$U = \text{ess inf}_{a \in \mathcal{A}} U_a.$$

By Lemma 2, a countable subfamily U_{a_n} determines U . Recursively choose c_n above c_{n-1} and a_n . Then U_{c_n} decreases to U modulo null sets, and

$$\lambda(U) = \lim_n \lambda(U_{c_n}) \geq \eta.$$

Again by Lemma 2, for each a choose a countable set C_a of large indices $b \geq a$ such that

$$U_a = \bigcup_{b \in C_a} E_b \quad \text{modulo null sets.}$$

Thus

$$D_a := U \setminus \bigcup_{b \in C_a} E_b$$

is null. Outside the fixed null set where $h_a \not\rightarrow 0$, every $x \in U$ has an index a_x after which $h_b(x) < \delta$. Consequently $x \notin E_b$ for $b \in C_{a_x}$, so $x \in D_{a_x}$. We have covered the positive-measure set U by one fixed null set and the family $(D_a)_{a \in \mathcal{A}}$. Lemma 1 gives $|\mathcal{A}| \geq \text{cov}(\mathcal{N})$. $\square$

5 The sharp lower bound

Theorem 4. *If $\mathcal{A}$ is a directed set with $|\mathcal{A}| < \text{cov}(\mathcal{N})$, then*

$$\mathcal{NM}_{\mathcal{A}} = \mathcal{NF}_{\mathcal{A}} = \mathcal{ND}_{\mathcal{A}} = \emptyset.$$

Proof. The dominated family. If $(f_a) \in \mathcal{ND}_{\mathcal{A}}$ has pointwise a.e. limit f and $|f_a| \leq g$ a.e., apply Lemma 3 to

$$h_a = |f_a - f|, \quad G = g + |f|.$$

This contradicts $|\mathcal{A}| < \text{cov}(\mathcal{N})$.

The monotone family. Suppose $(f_a) \in \mathcal{NM}_{\mathcal{A}}$ has pointwise a.e. limit f . Set

$$u = \text{ess sup}_{a \in \mathcal{A}} f_a.$$

Choose a_n determining this essential supremum and recursively choose c_n above c_{n-1} and a_n . The a.e. order makes (f_{c_n}) increasing modulo null sets, with supremum u . Ordinary monotone convergence yields

$$\int u = \sup_n \int f_{c_n} = \sup_{a \in \mathcal{A}} \int f_a. \quad (1)$$

We claim that $f = u$ a.e. Since $\{f_a > u\}$ is null for every a , pointwise convergence gives

$$\{f > u\} \subset N \cup \bigcup_{a \in \mathcal{A}} \{f_a > u\}, \quad (2)$$

where N is the null convergence-exception set. Lemma 1 and $|\mathcal{A}| < \text{cov}(\mathcal{N})$ imply $\lambda\{f > u\} = 0$.

For the reverse inequality, fix n . If $x \in \{f_{c_n} > f\} \setminus N$, convergence supplies an index after which the values are strictly below $f_{c_n}(x)$. Taking a common upper bound with c_n shows

$$\{f_{c_n} > f\} \subset N \cup \bigcup_{b \geq c_n} \{f_{c_n} > f_b\}. \quad (3)$$

Every set in the last union is null by the a.e. monotonicity. The same covering argument makes $\{f_{c_n} > f\}$ null. Taking the countable supremum gives $u \leq f$ a.e. Thus $u = f$ a.e., and (1) is precisely the monotone-convergence conclusion, a contradiction.

The Fatou family. Let (f_a) be a net of a.e.-nonnegative measurable functions. Define the literal and essential tail infima by

$$q_a(x) = \inf_{b \geq a} f_b(x), \quad g_a = \text{ess inf}_{b \geq a} f_b,$$

and put

$$f = \sup_{a \in \mathcal{A}} q_a = \liminf_{a \in \mathcal{A}} f_a, \quad g = \text{ess sup}_{a \in \mathcal{A}} g_a.$$

Assume, as in the definition of $\mathcal{NF}_{\mathcal{A}}$, that f is integrable.

For each a , a countable tail subfamily determines g_a , so $q_a \leq g_a \leq g$ outside a null set D_a . Therefore

$$\{f > g\} \subset \bigcup_{a \in \mathcal{A}} D_a.$$

The left side is measurable; Lemma 1 implies that it is null. Hence $f \leq g$ a.e.

The net (g_a) is increasing in the directed order. Countably determine its essential supremum and take successive common upper bounds as before. Ordinary monotone convergence then gives

$$\begin{aligned} \int f &\leq \int g = \sup_{a \in \mathcal{A}} \int g_a \\ &\leq \sup_{a \in \mathcal{A}} \inf_{b \geq a} \int f_b = \liminf_{a \in \mathcal{A}} \int f_a. \end{aligned}$$

This is Fatou's inequality and contradicts membership in $\mathcal{NF}_{\mathcal{A}}$. □

6 Consequences for Q1–Q3

Corollary 5 (Exact minimum in Q1 and Q2). *For every cardinal $1 \leq \mu \leq 2^{\mathfrak{c}}$, the smallest possible cardinality of a directed set in Q1 is*

$$\boxed{\text{cov}(\mathcal{N})}.$$

The same is true for Q2.

Proof. The lower bound follows from Theorem 4, since even one nonzero member of any requested cone would make the relevant family nonempty.

For the upper bound, Theorem 2.4 of the source paper [1] constructs in ZFC a directed set of cardinality $\text{cov}(\mathcal{N})$ and a positive cone of dimension $2^{\mathfrak{c}}$ contained simultaneously in $\mathcal{NM}_{\mathcal{A}}$, $\mathcal{NF}_{\mathcal{A}}$, and $\mathcal{ND}_{\mathcal{A}}$, after adjoining zero. Selecting μ generators gives every smaller requested dimension. For Q2, note additionally that $\mathcal{ND}_{\mathcal{A}}$ is symmetric under multiplication by -1 , so the same generators also give the negative cone required by the source's definition of coneability. $\square$

Corollary 6 (Exact minimum in Q3 through c). *For every cardinal $1 \leq \mu \leq \mathfrak{c}$, the smallest possible cardinality in Q3 is*

$$\boxed{\text{cov}(\mathcal{N})}.$$

This remains true if Q3 is read as requiring one common directed set for all three families.

Proof. The family $\mathcal{ND}_{\mathcal{A}}$ occurring in Q3 forces the lower bound by Theorem 4.

For the upper bound, the source constructs directed sets of cardinality $\text{cov}(\mathcal{N})$ on which, respectively, $\mathcal{AN}_{\mathcal{A}}$ is strongly $2^{\mathfrak{c}}$ -algebrable, $\mathcal{ND}_{\mathcal{A}}$ is strongly $\mathfrak{c}$ -algebrable, and $\mathcal{ANM}_{\mathcal{A}} \setminus \mathcal{ND}_{\mathcal{A}}$ is strongly $\mathfrak{c}$ -algebrable; see its Theorems 2.17, 2.13, and 2.8 [1]. Let these index sets be $\mathcal{A}_1, \mathcal{A}_2, \mathcal{A}_3$, and give

$$\mathcal{A} = \mathcal{A}_1 \times \mathcal{A}_2 \times \mathcal{A}_3$$

the coordinatewise order. Its cardinality is still $\text{cov}(\mathcal{N})$.

Pull a net on $\mathcal{A}_i$ back along the i -th coordinate projection. That projection is order preserving and cofinal, so the pointwise limit and every relevant eventual integral or convergence property are unchanged. The pullback is an injective algebra homomorphism, hence preserves freeness. Thus the three source algebras live on one common $\mathcal{A}$. Restricting their free generating sets gives every $1 \leq \mu \leq \mathfrak{c}$. $\square$

7 Verification, novelty, and limitations

The proof is structural and uses no numerical computation. It was audited for the two points most likely to hide a countability assumption: Lemma 2 explicitly reduces essential suprema and infima to sequences, while Lemma 3 never assumes a countable cofinal subset of $\mathcal{A}$. Equations (2) and (3) keep the varying a.e. exception sets visible rather than silently choosing common representatives.

A bounded novelty search through August 17, 2026 used the exact title, arXiv id, author, the phrase *smallest possible cardinality*, and combinations of $\text{cov}(\mathcal{N})$, nets, dominated convergence, monotone convergence, and lineability. It located the 2025 journal publication of the source and a later paper by Dantas–Rodríguez-Vidanes on different net pathologies, but no source stating Theorem 4 or Corollaries 5–6. Novelty is therefore plausible, not certified.

The result leaves open:

- the classification of all directed sets in Q1–Q3;
- Q3 for $\mathfrak{c} < \mu \leq 2^{\mathfrak{c}}$;
- the regular-cardinal Questions Q4–Q6.

The first remaining issue depends on the cofinal/Tukey structure of the index set, not only its cardinality. Above $\mathfrak{c}$, attempts to combine the source's maximal Bernstein coding with dominated-convergence failure meet a genuine cardinal-arithmetic bottleneck.

8 Human-review recommendation

The packet should receive mathematical review as a candidate substantial partial result. The priority checks are:

1. the passage from the down-directed tail essential unions to the positive-measure set U in Lemma 3;
2. the literal-versus-essential $\liminf$ comparison in the Fatou case;
3. the interpretation and cofinal pullback used for Q3.

References

- [1] D. L. Rodríguez-Vidanes, *Lineability on nets and uncountable sequences of functions in measure theory*, arXiv:2306.03862v1 (2023); published in Rev. R. Acad. Cienc. Exactas Fís. Nat. Ser. A Mat. 119 (2025), Article 35, DOI: 10.1007/s13398-024-01696-8.